
Pullbacks and pushouts

These are more advanced universal properties, showing more of the general features that were missing from the special cases we've seen so far.

The universal properties we have seen so far are terminal/initial objects, and products/coproducts. As we mentioned at the end of the last chapter, universal properties are all universal cones over some starting data. Our starting data so far has not been very general: it has not included any morphisms. For terminal/initial objects the starting data was trivial, that is, it was empty. For products/coproducts it was a pair of objects. In this chapter we are going to do a first example where the starting data is a diagram including some morphisms. This gives all the ideas needed for universal cones over general diagrams, although the formal definition of the full generality is beyond our scope.

Pullbacks and pushouts are dual to each other although neither of them is called “co” of the other. Pullbacks are also sometimes called “fibered products” especially in some fields where they really are fibered products. As this term indicates, they're a bit like products but restricted, or refined, in a sense we'll discuss.

This chapter is a culmination of our progressive exploration of universal properties. As these chapters have been a build-up, the material in this chapter is probably harder than what has gone before. However, the next chapter will begin a fresh topic, so if you feel confused in the current chapter I am optimistic that you will be able to pick up again in the next chapter, and come back to this one at a future point.

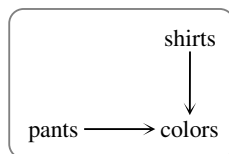
19.1 Pullbacks

Pullbacks can be thought of as refinements of products, intuitively as well as technically. Here is the intuitive idea for pullbacks.

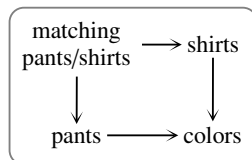
A product of two sets takes all the possible ways of making pairs, with one element from each set. But perhaps we want to make pairs that are coherent in some way, perhaps pairing up people who are the same height to be dancing

partners, or pairing pants/trousers with shirts that are the same color: we do that using a pullback.

If we were just pairing any old pants with any old shirts then our starting data would just be two sets: our set of pants, and our set of shirts. However, to match colors we need to take into account the functions shown on the right.



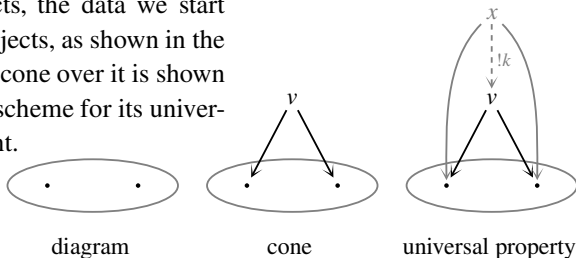
Then, we're looking for pants and shirts that are mapped to the same place in the set of colors, which gives us a commuting square as shown here.



But it's not just any old commuting square on the right, it's a *universal* one, giving all matching pants–shirts pairs, and nothing else.

This is an example of a *universal cone*. Terminal objects and products are also universal cones, but we didn't express those quite so explicitly. Here's how it works for products.

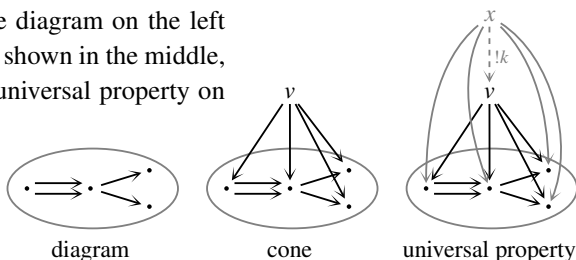
When we take products, the data we start with is just a pair of objects, as shown in the first diagram below. A cone over it is shown in the middle, and the scheme for its universal property on the right.



In a general limit, the starting diagram can be any diagram, involving arrows as well as objects.

For example, given the diagram on the left below, a cone over it is shown in the middle, and the scheme for a universal property on the right.

(The ellipse is just to make the cone look like a cone.)



A cone is an object v (the “vertex”) together with morphisms to every object in the starting diagram, such that everything commutes. To be universal it must then be universal among all cones of that shape: given any such cone with

vertex x , say, there must be a unique factorization $x \rightarrow v$, as shown in the third diagram. As before, being a factorization means it makes everything commute.

To express this rigorously in full generality much more formalism is needed, as we can't always simply draw these diagrams. Sometimes we want to take limits over infinite diagrams that thus can't be drawn. We will see how to express the notion of "diagram" formally in the next chapter using functors, but that's as far as we'll go. Here is the definition of a pullback, via universal cones.

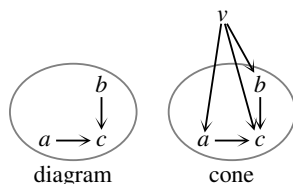
Definition 19.1 A *limit* over a diagram is a universal cone over it. A *pullback* in a category is a limit over a diagram of the shape shown on the right.



Admittedly we haven't exactly given a formal definition of "universal cone", only some pictures and the general idea. We'll now unravel what this means for the definition of pullback.

The first step is to understand what a cone over this diagram is. It will help us to name our objects, which I've done below.

A priori[†] a cone is then a vertex v together with morphisms to a , b and c making everything commute, as shown on the right.

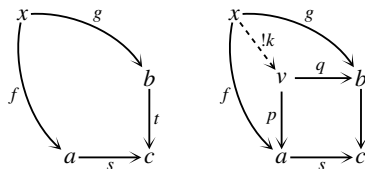


However, "everything commutes" means that the morphism $v \rightarrow c$ is determined by being the composite $v \rightarrow a \rightarrow c$, or indeed $v \rightarrow b \rightarrow c$ so it is redundant for us to specify it.

So this cone is actually just a commutative square — if we omit the arrow $v \rightarrow c$ and physically move v down, it becomes this:



The universal property then says: given any square[‡] with vertex x , say, as shown, there must be a unique factorization as shown on the right. I have named the arrows, for convenience.



Note that the conditions on being a factorization are the same as for a product — all the triangles involving projections must commute. I hope you will become happy with reading the condition from the diagram, but in algebra

[†] This means what we know in advance because of the general definition, before we sit down and think about the specifics of this particular situation.

[‡] As usual in category theory a shape with four arrows counts as a square even if it's not geometrically drawn as a square.

that condition says $[pk = f \text{ and } qk = g]$. No extra conditions arise from extra complications in the starting diagram; those only affect the notion of cone.

I have used some of the same notation here as for products, because this is a bit like a more restricted notion of product. The projections p and q are then a lot like the projections for a product.

To indicate a pullback we often put a little corner sign at the top left, as shown here. It goes where the vertex of the universal cone is.



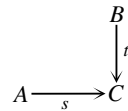
Definition 19.2 We say that a category *has pullbacks* when it has pullbacks over all diagrams of the relevant shape (that is, as shown on the right).



We will now look at some examples, as usual starting with sets and functions.

19.2 Pullbacks in Set

Given functions s and t as shown, we want to define a set which is like the canonical way to complete the diagram into a commuting square. It's going to turn out to be like a restricted product, so we will denote it $A \times_C B$.

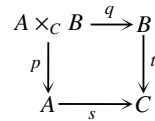


The idea is that if we didn't have the morphisms to C , the canonical thing to do would be to take the product $A \times B$, which is the set of all ordered pairs. However, to make sure the diagram commutes we take the “fibered” product, which means we restrict to those ordered pairs (a, b) such that a and b are mapped to the same place in C . This is a subset of the product $A \times B$.

Formally we define this: $A \times_C B = \{(a, b) \in A \times B \mid s(a) = t(b)\}$.

As this is a subset of the product $A \times B$, it inherits projection functions onto the A component and B component respectively.

Proposition 19.3 *The square on the right is a pullback, where p and q are the projections inherited from $A \times B$.*

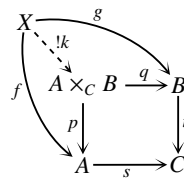


Things To Think About

T 19.1 See if you can prove this for yourself. Try the method for products of sets in the previous chapter, constructing a factorization by following through the definitions — there is no choice of what to do here.

When I say there's “no choice” of what to do, I mean there aren't really any decisions to make. Sometimes category theory proofs are like an automatic conveyor belt that carries you along if you can just manage to step on it.

Proof Consider a commuting diagram of sets and functions as in the outside of this diagram. We need to show there is a unique factorization as shown.



We need to say what $k(x)$ is for each $x \in X$. It has to be an ordered pair, and the commutativity of the two triangles means the A component has to be $f(x)$ and the B component has to be $g(x)$, just like for the product; we then have to check that's a valid element of $A \times_C B$. Again, this does uniqueness first, and existence afterwards.

Commutativity of the triangles means that if a factorization k exists, it must be given by $k(x) = (f(x), g(x))$. (This is called “unique by construction”.)

We need to check that this is an element of $A \times_C B$, that is, that the two components map to the same element of C . This follows by the commutativity of the outside of the diagram (see below). Thus k is indeed a factorization and is unique by construction, so the diagram is a pullback as claimed. $\square$

I hope that last part will make sense to you if you follow things around the diagram. Written in algebra, the condition we need to check is $sf(x) = tg(x)$ which is exactly the commutativity of the outside.

Note that when we write the vertex of the pullback as $A \times_C B$ it only tells us what the corner object C is, without reference to the morphisms s and t . In some situations those are very natural, but I think it's good to be aware that the notation is missing something.

Note on terminology

Sometimes we talk about pulling back morphisms along each other. So we might say that p is the pullback of t along s , and that q is the pullback of s along t . Although the definition is symmetrical in s and t sometimes the situation isn't. Sometimes one of the starting morphisms is more interesting and the other is in more of a supporting role.

Things To Think About

- T 19.2** 1. See if you can show that there is always a canonical morphism from the pullback $a \times_c b$ to the product $a \times b$, if they exist. (Use the universal property of the product to induce it.)
2. Show that this canonical morphism is always monic. This shows that the pullback is always something like a restricted version of a product.
3. Show that if c is the terminal object the pullback $a \times_c b$ is a product $a \times b$.

We won't rely on those thoughts for anything, so I'm just going to leave them there. Before we move on, I do want to show you my favorite square.

The diagram on the right is officially my favorite square, for reasons we're about to investigate. Here every arrow is the subset inclusion function.[†]

$$\begin{array}{ccc} A \cap B & \longrightarrow & B \\ \downarrow & & \downarrow \\ A & \longrightarrow & A \cup B \end{array}$$

Things To Think About

T 19.3 See if you can show that the above square is a pullback in **Set**. (This is half of why it's my favorite square.)

We know from our previous construction that the pullback should consist of pairs $(a, b) \in A \times B$ such that a and b land in the same place in $A \cup B$. But the only way for elements a and b to do that in this case is if they are the same, and in the intersection.

Note that this is not exactly a rigorous proof, but if we turned it into one we still wouldn't have directly checked the universal property of the square shown: we'd have shown that the diagram shown is suitably isomorphic to the construction of the pullback we did earlier. At that point we'd need to invoke a theorem saying that everything isomorphic to a pullback is a pullback. I hope you can *feel* that this is true, from the sense we've developed of universal properties so far. To be sure it's true, in the most abstractly satisfying way, we will now re-frame pullbacks as terminal objects in a more complicated category related to $\mathcal{C}$.

19.3 Pullbacks as terminal objects somewhere

We have said that all universal properties are terminal objects *somewhere*, and showed that products in $\mathcal{C}$ are terminal objects in a category “souped-up” from $\mathcal{C}$. We can do this for pullbacks too. The idea is as for products: we start with a diagram and want to find a universal cone for it, so we make a category of all cones, and the terminal objects in there are the universal cones.

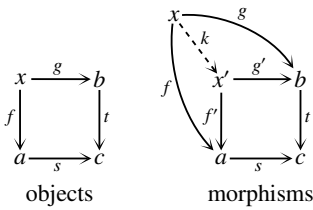
Things To Think About

T 19.4 Suppose we want to define pullbacks for the diagram shown on the right. See if you can make a “souped-up” category of cones over the diagram, in which terminal objects are pullbacks for the diagram.

$$\begin{array}{ccc} & & b \\ & & \downarrow \\ a & \longrightarrow & c \end{array}$$

[†] Remember $A \cap B$ is the intersection, that is, the set of all objects in both A and B , and $A \cup B$ is the union, where you take all the elements of A and all the elements of B and throw them in a set together (and if anything is in both sets you just take it once).

We make a souped-up category as follows. Objects are commutative squares as shown, and morphisms are given by a morphism k making the diagram on the right commute. Terminal objects in this category are then pullbacks over the original diagram.



Note that for products we started with just the objects a and b , without the object c and morphisms s and t . If you erase c, s and t in the above diagrams, you get back the souped-up category for defining products; this new category just has an extra condition of commutativity with s and t .

Things To Think About

T 19.5 What can we now deduce immediately, analogous to the things we deduced for products, from the fact that this is a terminal object somewhere?

We can now immediately deduce that pullbacks are unique up to unique isomorphism, where the isomorphism in question is unique making the whole diagram commute as in the diagram for a factorization.

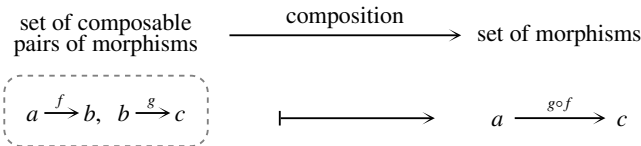
19.4 Example: Definition of category using pullbacks

Before we move on to the dual of pullbacks I'd like to take a short aside to show one important use of pullbacks: to finish the alternative definition of category that we started talking about in Section 17.5.

This approach started with a set C_1 of morphisms and a set C_0 of objects, equipped with functions giving the source and target of each morphism as shown on the right.

$$C_1 \begin{matrix} \xrightarrow{s} \\ \xrightarrow{t} \end{matrix} C_0$$

We showed how to express identities in this context, but we couldn't do composition because it needs pullbacks. The idea is that composition is a function:



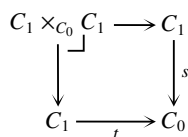
A “composable pair” of morphisms looks like an ordered pair (f, g) but it's not *any old* ordered pair: the two morphisms have to meet in the middle in order to be composable. That is, the target of f has to be the same as the source of g .

Things To Think About

T 19.6 See if you can express that as a pullback in **Set**.

We take the pullback shown on the right; it's a pullback in **Set** so can be defined as follows:

$$C_1 \times_{C_0} C_1 = \{ (f, g) \in C_1 \times C_1 \mid t(f) = s(g) \}$$



That is, an element is a pair of morphisms of the category we're defining, where the target of one is the source of the other: a composable pair.[†]

Composition is then a function $C_1 \times_{C_0} C_1 \longrightarrow C_1$ with some conditions.

Note on binary operations

The fact that composition is defined on the pullback rather than the product is what makes it a *partial* binary operation rather than a binary operation: it is partial as it is only defined on part of the set of ordered pairs.

A binary operation on a set A is a function $A \times A \longrightarrow A$.

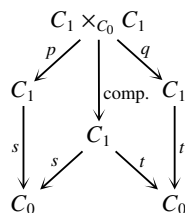
Our function for composition is defined on a pullback, which is a subset of the product, not the whole product.

We still need to make sure that the composition function gives something that behaves like composition. We are aiming for this: $a \xrightarrow{f} b \xrightarrow{g} c = a \xrightarrow{g \circ f} c$.

First we have to make sure that the source and target of the composite are correct, that is, that the conditions on the right hold.

$$\begin{aligned} s(g \circ f) &= s(f) \\ t(g \circ f) &= t(g) \end{aligned}$$

We can express this by the commutative diagram on the right. Note that s and t appear twice each in the equations, so they appear twice each in the diagram too.

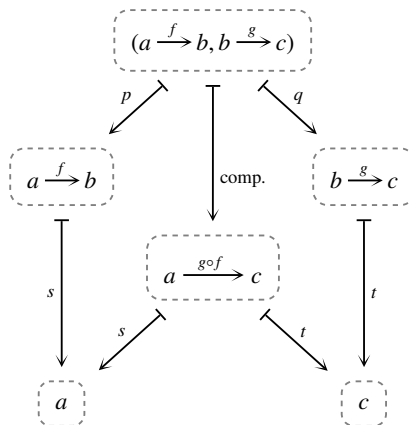


Things To Think About

T 19.7 See if you can “chase” a composable pair (f, g) around this diagram to see how it gives the condition we want. It can sometimes be helpful to draw the action on elements around the outside of the original diagram, or if that makes a mess, draw a separate diagram with the same shape, but with elements at the vertices instead of the sets C_0 , C_1 and so on.

[†] Note that $t(f)$ isn't a composite; it's the result of applying the function t to the element $f \in C_1$.

Here is the diagram chase; I hope you will gradually become used to doing this for yourself so that the commutative diagram is enough. Eventually you may find that you don't even need to write out the diagram to chase elements around, but you can just do it by following the commutative diagram around in your head.



Things To Think About

T 19.8 See if you can now express the axioms for associativity and identities in commutative diagrams.

What we have started to do here is express the definition of a category categorically, if that doesn't sound too self-referential. That is, we have expressed it entirely using objects and morphisms in the category **Set**. One big payoff for having done that is that we can then pick it up and put it in other categories and see what happens. This gives us the notion of *internal category* and we'll come back to it in the last chapter, on higher dimensions.

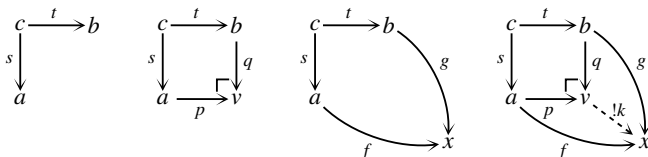
19.5 Dually: pushouts

We will now move on to the dual of a pullback, which is called a pushout.

Things To Think About

T 19.9 Write down the dual concept of a pullback. That is, if we have a pullback in $\mathcal{C}^{\text{op}}$ what does this translate to as a structure in $\mathcal{C}$?

Definition 19.4 A *pushout* is a universal cocone for a diagram of the shape shown in the first diagram below. That is, it is a vertex v and morphisms as shown in the second diagram, such that given any diagram as in the third diagram, there is a unique factorization as shown in the fourth diagram.



We indicate a pushout using the notation shown: a small corner marking the vertex of the universal cocone.

Note that as this is a cocone the factorization goes *out* of the universal vertex. In a cocone we have coprojections, that is, morphisms from the diagram in question to the vertex. Then the direction of the factorization sorts itself out by type-checking — it is a factorization of the non-universal cocone, a way of expressing the non-universal one as a composite of the universal one and the factorization morphism. The arrow can then only go one way.

Things To Think About

T 19.10 What do we immediately know about pushouts, dually to what we know about pullbacks?

Dually to pullbacks we immediately know the following things:

- Pushouts are initial objects somewhere.
- Pushouts are unique up to unique isomorphism.
- Pushouts are canonically related to coproducts.

19.6 Pushouts in **Set**

Recall from Section 19.2 that I said the square on the right is my “favorite square”. We have so far seen part of the reason; we can now see all of it.

$$\begin{array}{ccc} A \cap B & \longrightarrow & B \\ \downarrow & & \downarrow \\ A & \longrightarrow & A \cup B \end{array}$$

Things To Think About

T 19.11 We showed that this is a pullback; can you now show that it is a pushout as well?

Proof We need to exhibit the universal property, as in the diagram on the right. So we start by considering the outside, that is, a set X together with functions f and g making the outside commute.

$$\begin{array}{ccccc} A \cap B & \xrightarrow{q} & B & & \\ p \downarrow & & \downarrow t & \searrow g & \\ A & \xrightarrow{s} & A \cup B & \xrightarrow{!k} & X \\ & \searrow f & & & \end{array}$$

As for when we did the pullback property, it may help to think about what it means for the outside to commute. If we start with an element $y \in A \cap B$ and chase it around, we see that commutativity means $f(y) = g(y)$, that is the functions f and g have to agree on the intersection $A \cap B$.

We need to construct a unique factorization. Now, the commuting condition for a factorization k amounts to the conditions on the right (since s and t are subset inclusions).

$$\begin{aligned} \forall a \in A, \quad k(a) &= f(a) \\ \forall b \in B, \quad k(b) &= g(b) \end{aligned}$$

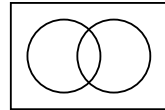
The question now is: can we use this as a *definition* of k ? When defining a function on a union it works to define it on the individual parts as long as the definition agrees on the intersection, and that's true here as we know f and g agree on the intersection. We refer to this as the function being “well-defined”.[†] We just need to say that formally.

So we define k by cases as shown on the right; k is well-defined because the outside of the diagram commutes, so if $z \in A \cap B$ then $f(z) = g(z)$ as needed.

$$k(z) = \begin{cases} f(z) & \text{if } z \in A \\ g(z) & \text{if } z \in B \end{cases}$$

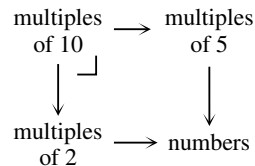
Thus we have defined a factorization k and it is unique by construction. $\square$

This is my favorite square because it is a pullback *and* a pushout, and it encapsulates some very fundamental principles in a way that I find much more structurally compelling than the usual Venn diagram (although I concede that the Venn diagram is better for actually putting things in).



One thing we can do with the pullback notation is stress that something that is not *a priori*[‡] the intersection is in fact isomorphic to the intersection.

For example, we can state that the commuting square of subset inclusions on the right is a pullback. The content of that statement is that the multiples of 10 are *precisely* those numbers that are multiples of both 2 and 5.

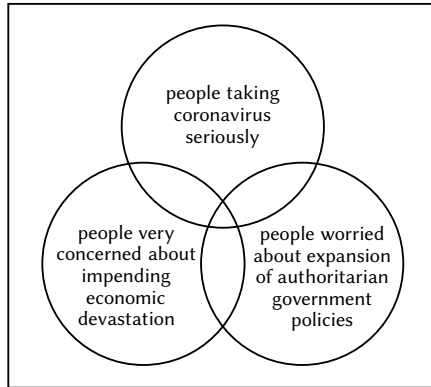


[†] This doesn't just mean we've defined it well, it means we haven't created an ambiguity in the definition, which is what would happen if our definition on individual parts did not agree on the intersection.

[‡] This means that we don't know in advance that it's the intersection, as we didn't define it to be the intersection, but we can then deduce that it is (isomorphic to) the intersection.

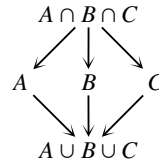
I saw a Venn diagram meme yesterday that showed the diagram on the right and said “Believe it or not, it’s OK to be in all three.”

I would go a step further and say that the intersection of all three *exactly* consists of sensible people. That is, that all sensible people are in the intersection, and that everyone in the intersection is sensible.



Well, I admit that the last part might be more of a stretch (it’s probably possible to be sensible about these issues and perhaps not at all sensible about something completely unrelated).

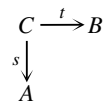
But if I really did think that the intersection was precisely the sensible people, I could express it as a universal property. It would be a 3-fold generalization of the pullback/pushout square above, and might look something like the diagram on the right.



Incidentally I think it is important to note that people define “authoritarian government policies” in different ways, and if someone doesn’t think an illness is really dangerous, they might well think that a drive to vaccinate everyone is authoritarian, rather than protective.

Anyway these intersection/union situations are a particular kind of pushout in **Set**, and as we did with pullbacks we will now look at more general pushouts in **Set**. The idea is that the union is a way of sticking together two sets where we match them up along the elements they have in common. We could stick together two sets even if they have no elements in common, by defining the place we’re going to match them up ourselves. This is how general pushouts work in **Set** and is actually quite a lot harder to define than pullbacks.

Consider the diagram of sets and functions on the right. To make a cocone we need to stick A and B together in such a way that C ends up in the same place (that’s the commutativity).



To make it universal we have to do it in the most economical way possible, that is, we shouldn’t stick anything together that doesn’t need to be stuck together.

Things To Think About

T 19.12 Try these two examples of sets and functions as shown. In each case make a set that is like sticking A and B together (union) but where you match up $s(c)$ with $t(c)$ everywhere, for each $c \in C$.

$$\begin{array}{ccc} C & \xrightarrow{t} & B \\ s \downarrow & & \\ A & & \end{array}$$

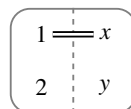
$$\begin{array}{lcl} 1. & A = \{1, 2\} & \\ & B = \{x, y\} & \\ & C = \{c\} & \end{array} \quad \begin{array}{ccc} & s & t \\ c \mapsto 1 & & c \mapsto x \end{array}$$

$$\begin{array}{lcl} 2. & A = \{1, 2\} & \\ & B = \{x, y\} & \\ & C = \{c, d\} & \end{array} \quad \begin{array}{ccc} & s & t \\ c \mapsto 1 & & c \mapsto x \\ d \mapsto 2 & & d \mapsto x \end{array}$$

The first example is not too unlike the intersection/union situation; in fact, it is isomorphic to an intersection scenario as the morphisms s and t are injective. If we re-named the elements then it would be just like an intersection/union situation. For example we could rename the elements of B to be 2 and 3 and then the element of C could be 2.

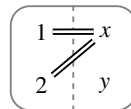
If we don't do that, we need to make a new set by starting with the disjoint union $A \sqcup B$ and then "identifying" $s(c)$ with $t(c)$ to ensure that the putative pushout square will commute. That is, we identify 1 with x . This means that we stick them together, or declare them to be the same object.

We could picture it as shown on the right. The large box with the dotted line represents $A \sqcup B$, with A on the left and B on the right. The long equals sign shows that we have identified the elements 1 and x .



So the resulting set has 3 elements: $1=x$, 2, and y .

The second one is more complicated. We need to identify $s(c)$ with $t(c)$, so we identify 1 and x . We also identify $s(d)$ with $t(d)$, that is, 2 and x . This gives the picture here.



We now see that, as a consequence, 1 and 2 also end up being identified with each other. So the resulting set has only two elements: $1=x=2$, and y .

This process is called *quotienting out by an equivalence relation*. Technically, however, the relation we're starting with isn't necessarily an equivalence relation, so actually we first have to use it to *generate* an equivalence relation, and then we quotient out by it. In my informal argument above, the relation we start with is the thing I've drawn with long equals signs. The fact that 1 and 2 became identified via x was not part of the original relation, but was a consequence of generating an equivalence relation.

We start with the following relation on $A \sqcup B$: $s(z) \sim t(z) \quad \forall z \in C$

Things To Think About

T 19.13 In what ways does this relation fail to be an equivalence relation in the above two examples? (Assume that the definition is symmetric.)

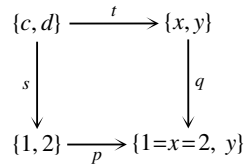
This relation is definitely not reflexive, as nothing is related to itself; however, it's easy to add that condition into a relation. We can also straightforwardly define it to be symmetric. Transitivity is more interesting: we have $1 \sim x$ and $x \sim 2$, but we don't know in advance that $1 \sim 2$, so transitivity fails.

Generating an equivalence relation consists of taking the “smallest” equivalence relation that satisfies the relation we already have. In this case it essentially consists of “closing” the relation under transitivity.

For the pushouts of sets we're not so concerned with what that equivalence relation is, as much as with what set we get when we quotient out by it. Quotienting out by a relation is where we identify everything that is related. Technically there are various ways to do this[†] but they're all isomorphic as sets, and so from the point of view of pushouts it really doesn't matter which one we do as the category will treat them all as “the same”.

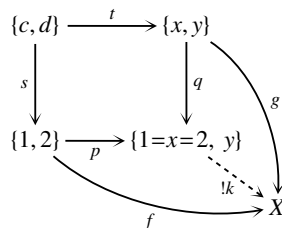
Things To Think About

T 19.14 Can you show that our second construction above (depicted on the right) is a pushout in **Set**? Remember that “ $1=x=2$ ” is a single element, so the set at the bottom right has only two elements. The coprojections p and q send everything to itself, wherever it appears in that weird-looking set.



I think the important thing here is the idea: we made the set at the bottom right-hand corner by sticking elements together as economically as possible to make a commuting square. So it “should” be universal. However, for it to be rigorous mathematics we must also establish this formally.

Proof We need to exhibit the universal property, as shown on the right. So we start by considering the outside, that is, a set X together with functions f and g making the outside commute. We need to construct a unique factorization k . (Again, the $!$ symbol signifies uniqueness.)



[†] If you're interested you could look up “equivalence classes” or “representatives of equivalence classes”.

The conditions for a factorization say that the two triangles in the above diagram must commute. The right-hand one tells us k must act as shown on the right.

$$\begin{array}{ccc} & & k \\ 1=x=2 & \mapsto & g(x) \\ y & \mapsto & g(y) \end{array}$$

We must now check that the left-hand triangle commutes; this says we must have $k(1=x=2) = f(1) = f(2)$.[†]

This is true because the outside of the diagram commutes, so in particular

$$\begin{aligned} f(1) &= fs(c) && \text{by definition of } s \\ &= gt(c) && \text{by commutativity} \\ &= gt(d) && \text{by definition of } t \\ &= fs(d) && \text{by commutativity} \\ &= f(2) \end{aligned}$$

Note that I worked this out helped by deep belief that the quotient on the set in the bottom right is the exact right thing to make this pushout work, and thus that whatever we need to be true must be true. Believing that something is true before proving it can be dangerous as it can result in leaps of faith rather than rigor, but if the belief comes from deep structural understanding it can guide us towards rigor.

So the above definition of k makes the diagram commute and is unique by construction, so the square is a pushout as claimed. $\square$

I honestly think that was quite ugly and in fact the proof in generality makes more sense.[‡]

But I would also like to point out that these things just *are* a bit ugly. Quotienting out by relations is an ugly construction. We're sort of forcing things to become the same as each other and it is typically a messy thing to do because it comes with consequences. The more structure we have around, the more consequences we end up with, and the messier it gets. Here we don't have any structure, just elements, and it's already pretty messy.

Here's the construction of pushouts for general sets.

[†] If this confuses you, “chase” the elements 1 and 2 round the diagram individually, and write down what it means for each one to land in the same place in X by both routes.

[‡] Actually with some more advanced understanding of category theory you'll see that the example is quite contrived. The function s is an isomorphism and some general theory then tells us the pushout square is trivial: q is also an isomorphism and p does essentially “the same” as t . I still think it's a good exercise to examine the construction in a very small example.

Proposition 19.5 *The diagram on the right has a pushout given by $A \sqcup B / \sim$ where the relation $\sim$ is generated by:*

$$s(c) \sim t(c) \quad \forall c \in C$$

$$\begin{array}{ccc} C & \xrightarrow{t} & B \\ s \downarrow & & \\ A & & \end{array}$$

The notation $A \sqcup B / \sim$ means we are taking the disjoint union and quotienting out by the relation given.

Note that this construction makes the relationship with the coproduct explicit: we in fact start by taking the coproduct and then we quotient it. That quotient map is a function $A \sqcup B \rightarrow A \sqcup B / \sim$ which is in fact the unique factorization induced by the universal property of the coproduct.

Note on defining a function on a quotient

At one level this proof is not going to be completely rigorous as we haven't really defined this quotient completely rigorously. However, really the one thing you need to understand about quotients is how we define functions *out* of them. We are essentially going to be defining a function out of a quotient, like this:

$$V / \sim \longrightarrow X.$$

The quotient $V / \sim$ consists of the elements of V at root, but some of them will be identified under the relation $\sim$. To define a function to X we can start by simply defining a function on *all* the elements of V , that is, a function $V \xrightarrow{k} X$.

We then show it is *well-defined*, that is, that it respects the relation $\sim$ so that if we then use this function on the quotient instead of on V no weird incompatibilities will happen. More precisely, we need to check that related elements in V are mapped to the same place by k , that is $v \sim v' \implies k(v) = k(v')$.

This works whether we start with an equivalence relation, or a relation that we will then use to generate an equivalence relation.

Proof We need to exhibit the universal property, as shown on the right. So we start by considering the outside, that is, a set X together with functions f and g making the outside commute. We need to construct a unique factorization k .

$$\begin{array}{ccccc} C & & \xrightarrow{t} & & B \\ s \downarrow & & & & \downarrow q \\ A & \xrightarrow{p} & A \sqcup B / \sim & \xrightarrow{!k} & X \\ & \searrow f & & & \uparrow g \end{array}$$

The conditions for a factorization tell us that the conditions on the right must hold, so we define a function k by those conditions.

$$\begin{array}{ll} \forall a \in A & k(a) = f(a) \\ \forall b \in B & k(b) = g(b) \end{array}$$

We must show that this respects the relation $\sim$ (generated by $s(c) \sim t(c)$ for all $c \in C$). Now, given $c \in C$:

- $s(c) \in A$ so $ks(c) = fs(c)$ by definition of k .
- $t(c) \in B$ so $kt(c) = gt(c)$ by definition of k .
- The outer diagram commutes so $fs(c) = gt(c)$.

Thus k does respect the relation, so defines a function on the quotient.[†] Thus we have defined a factorization and it is unique by construction. $\square$

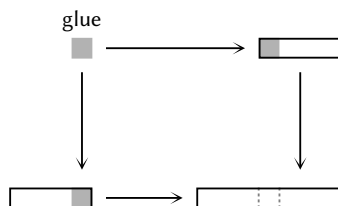
Things To Think About

T 19.15 Go back to our proof of the universal property for the specific case (T 19.14) and see if you can see how that proof relates to our general one above.

19.7 Pushouts in topology

Pushouts in spaces are a key way of glueing spaces together to make more complex spaces. The idea is that when we (physically) glue things together we choose an area on each thing where the glue will go, and then those areas get identified.

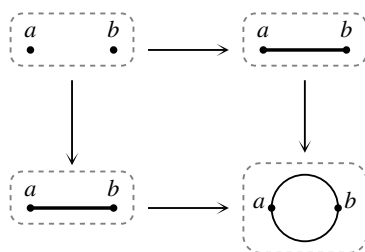
For example here's a pushout square glueing two strips of paper together to make a longer strip. It's an example of my "favorite square" but this time in the category **Top** of topological spaces and continuous maps.



General pushouts are again more complicated, but these easier ones are extremely useful for building up spaces in topology by sticking them together.

[†] You might prefer this sentence to say "Thus k does respect the relation, so it defines a function on the quotient". Mathematical writing often omits the "it" though. It streamlines the sentence, but it might just be a habit from the old days of desperately saving space in print journals.

The above example of making a longer strip was not that interesting but here is a pushout that sticks two intervals together to make a circle. The two copies of the interval end up on one half of the circle each, and the endpoints get glued together.



Remember that we are working in topology so the *geometry* of the situation doesn't matter. That is, it doesn't matter that the straight lines become curved; all that matters is how they are attached.

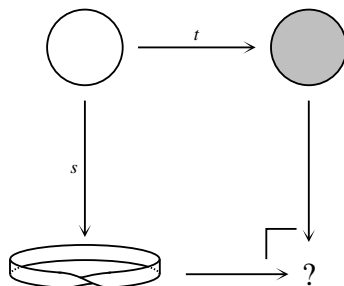
Pushout constructions in **Top** enable us to construct things that are very difficult to picture. For example, we could consider glueing a disk (a filled-in circle) to the edge of a Möbius strip.

A Möbius strip is the shape produced by taking a strip of paper and glueing the ends together with a twist.



This shape famously has “only one side” as the front and back of the paper have been joined up. It also only has one edge, because the “right and left” edges of the strip of paper have been joined up. As it only has one edge, this means its edge (or boundary) is topologically a circle. Thus we can abstractly (though not physically) glue a disk onto that boundary.

The square on the right is the pushout in question, where the morphisms s and t are each the inclusion of the circle into the boundary of the target shape. The shading is to show that the top right shape is a solid disk, whereas the top left shape is just the boundary and a hole in the middle. (I haven't shaded the Möbius strip as it was too hard.)



This is extremely hard to imagine visually or physically, but it is one possible construction of a very interesting space called the projective plane. In fact, many interesting spaces can be made by cutting disks out of things and glueing a Möbius strip into the circular hole that was made. This helps with the *classification of surfaces*, where all possible 2-dimensional surfaces are organized up to homotopy equivalence. All the so-called non-orientable surfaces can be made from a sphere by cutting out a finite number of holes and glueing

a Möbius strip onto each one. (The orientable ones are all the toruses with a finite number of doughnut-holes.)

19.8 Further topics

Pushouts of groups

We already saw that coproducts are much more complicated for monoids and groups than for sets and posets. This is because of the binary operation: it means that after making the disjoint union of the underlying sets, we must then generate new elements produced by the binary operation.

To find pushouts of groups we have to start with the coproduct of groups, which we saw in the previous chapter is the “free product”, and then we have to do something analogous to quotienting out by an equivalence relation. In group theory this is provided by the theory of quotient groups. The idea is that one can “divide” a group by a special kind of subgroup, called a normal subgroup. We have actually seen an example of this although we didn’t put it that way: the integers modulo n can be constructed as a quotient group where you start with the integers and then quotient by the subgroup consisting of all the multiples of n .

Pushouts of groups are related to those of topological spaces in a key way.

Pushouts in algebraic topology

Algebraic topology is about making relationships between spaces and groups. One way is via a functor that we’ll mention in the next chapter, which assigns to every topological space its *fundamental group*. This gives us a way of studying spaces via the rich and well-established theory of groups.

A big problem is then actually calculating what the fundamental group is as a group, rather than just as an abstract definition. Pushouts help us do this because of a wonderful theorem called Van Kampen’s theorem. This tells us that under certain circumstances we can express a space as a pushout and then calculate its fundamental group as a pushout of groups. The circumstances are a little stringent and, for example, would not help us in the situation above where we glued two intervals together to make a circle.

The pushout method works much better if instead of using groups we use groupoids. Recall that these are a “one dimension up” version of groups: groups are categories with a single object and every morphism invertible, whereas groupoids have every morphism invertible but no restriction on the objects.

In some ways the correspondence between spaces and groupoids works better than the one with groups, but there is much less long-standing groupoid

theory than there is group theory. Thus there is still great benefit to making the effort to produce a group rather than a groupoid. However, as we move into higher dimensions the balance of advantage starts to shift, as we will see later.

More general universal properties

We have hinted at a more general theory of universal properties as we have built up more complicated versions:

1. We started with terminal and initial objects. These are universal (co)cones over empty diagrams: no objects and no morphisms.
2. We then saw products and coproducts. These are universal (co)cones over discrete diagrams: no morphisms.
3. Finally we saw pullbacks and pushouts. These are universal (co)cones over diagrams with just two morphisms pointing in opposite directions.

Another fairly straightforward type of universal property that we might have done next is equalizers and coequalizers.

These are universal (co)cones over a parallel pair of arrows, as shown here. You might like to try unraveling those definitions.

$$\cdot \rightrightarrows \cdot$$

In general, universal cones over a diagram are called *limits* and universal cocones are called *colimits*, so all the universal properties we've been seeing have been limits and colimits. Pullbacks and pushouts are a good prototype for how limits and colimits generally work. In fact all finite limits (that is, limits over finite diagrams) can be constructed from combining finite products and pullbacks.

This result is usually stated in terms of products and equalizers; I just said it with pullbacks as we haven't really looked at equalizers. Equalizers and pullbacks are interchangeable in the presence of products, that is, we can make equalizers out of pullbacks and products, and we can also make pullbacks out of equalizers and products. It is perhaps more intuitive to state the result about limits in terms of products and equalizers, as products deal with multiple objects and equalizers deal with multiple morphisms between the same objects, and that is really the two features that a diagram can have.

In the next chapter we will, among other things, see how to express general diagrams other than just by drawing them.